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FLOW FIELD AND STRUCTURE OF TURBULENT HIGH-PRESSURE PREMIXED METHANE/AIR FLAMES

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ABSTRACT

The present study focuses on the flow field characterization of highly turbulent premixed flames, typical for stationary gas turbines. Mean flame front position and flame front structure at high inlet temperatures, lean mixtures, and high pressures are studied, too. Turbulence intensities and integral length scales have been measured in an isothermal flow field with the help of Particle Image Velocimetry (PIV). Mean flame front position and flame structure have been studied using Planar Laser-Induced Fluorescence (PLIF) of the OH radical.

Turbulence intensities and integral length scales have been measured for different turbulence generating grid geometries and operating conditions. The results show that the combustor flow field can be divided in a region close to the combustor head, where grid-generated turbulence is dominant, and a region further downstream, strongly influenced by turbulence generated in the shear layer. In general the measured turbulence intensity scales well with the bulk velocity.

For a systematic variation of the turbulent Reynolds number, Damköhler number, and Karlovitz number the mean flame front position and the flame front structure were investigated. Increasing the pressure and thereby mainly increasing the turbulent Reynolds number only slightly affects the mean flame front position but increasingly corrugates the flame front. Increasing the bulk velocity and thereby the turbulence intensity does not affect the mean flame front position but due to the higher turbulence the flame front is increasingly corrugated.

Keywords:
High Pressure Lean Premixed Flames, Turbulence, Flow Field, Flame Structure

INTRODUCTION

Turbulent premixed flames are widely spread in technical applications and are used especially in stationary gas turbines for a high-efficient and low emission energy conversion of gaseous fuels. Although lean premixed combustion of gaseous fuels is a well-established technique for power generation, there is still not sufficient knowledge of turbulence – chemistry interaction. Especially high-pressure experimental data at gas turbine relevant conditions is missing.

Turbulent premixed flames can be classified in different flame regimes with the help of the so-called Borghi diagram, presented in Figure 1 [1].

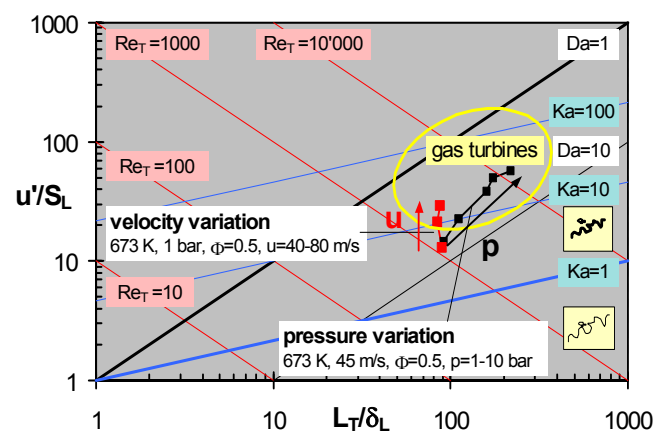

Figure 1: Borghi diagram and studied flames

In this diagram the characteristic parameters, turbulent Reynolds number Re_T, Damköhler number Da, and Karlovitz number Ka are plotted as a function of the characteristic turbulence parameters, turbulence intensity u' and integral

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length scale L_T, both being normalized with the corresponding characteristic parameters of laminar premixed flames (laminar flame speed S_L and flame thickness δ_L). In the Borghi diagram different flame regimes, which can be distinguished with respect to the different values of these parameters are shown.

For scaling purposes it is useful to set the Schmidt number $Sc = \nu/D$ equal to unity. Therefore the turbulent Reynolds number, the Damköhler number, and the Karlovitz number can be defined as

$$Re_T = \frac{u' \cdot L_T}{\nu} = \frac{u'}{S_L} \cdot \frac{L_T}{\delta_L}, \; Da = \frac{L_T}{\delta_L} \cdot \left(\frac{u'}{S_L}\right)^{-1}, \quad Ka = \left(\frac{u'}{S_L}\right)^{\frac{3}{2}} \cdot \left(\frac{L_T}{\delta_L}\right)^{-\frac{1}{2}}, \tag{1}$$

where ν is the kinematic viscosity.

Typical flames in stationary gas turbines are characterized by extremely high turbulent Reynolds numbers ($Re_T > 10^4$), Damköhler numbers about 1, and Karlovitz numbers higher than 10. Therefore these flames belong to the so-called *thickened turbulent flames* regime. According to theory the flame front in this regime is thicker than the sizes of the smallest eddies, therefore these eddies can penetrate into the flame front and thereby influence the reaction rate. In the last years Peters [2, 3] has proposed a new theory for the regime $Re_T > 1$ and $1 < Ka < 100$. In this theory, the flame front in the *thin reaction zones* regime is characterized by a thick preheating zone and a thin inner reaction layer, which is thinner than the sizes of the smallest eddies so they cannot penetrate into this inner layer.

A number of experimental and theoretical studies dealing with flame structure and turbulent flame speed of turbulent lean premixed flames can be found in the literature, but most of them are limited to atmospheric pressure, e.g. [4, 5]. Because of the influence of pressure on the characteristic parameters, especially on the turbulent Reynolds number, these results cannot be transferred to high-pressure gas turbine combustion. Other studies e.g. [6] were performed at higher pressures but either in SI engines, in which the turbulence intensities and the turbulence scales change during the compression period, or in constant volume vessels, in which the pressure change during the flame propagation is inherent, or the operating conditions were at near stoichiometric equivalence ratios [7]. Therefore again all these results are not representative for gas turbine combustion.

This was the motivation for the present study of lean premixed flames at typical gas turbine conditions. The characteristic parameters listed in Table 2 were calculated using a correlation for the laminar flame speed S_L obtained from a numerical study [8]. The laminar flame thickness δ_L has been calculated using the quotient of mass diffusion coefficient of methane in air D and the laminar flame speed. The turbulence intensity u' and the integral length scale L_T were both measured with the help of PIV. In the calculation of the turbulence intensity u' the equity of the root mean square deviation of the radial and circumferential velocity had been assumed. The turbulence intensity can than be calculated with:

$$u' = \sqrt{\frac{u(\mathrm{RMS})^2 + 2v(\mathrm{RMS})^2}{3}} \tag{2}$$

The integral length scale L_T has been calculated with the following equations, in which R_L is the correlation coefficient and r the correlation length:

$$R_L(r) = \frac{\overline{\Delta u(x) \cdot \Delta u(x+r)}}{u'(x) \cdot u'(x+r)} \tag{3}$$

$$L_T = \int_0^\infty R_L(r)\,dr \tag{4}$$

In the calculation of the characteristic parameters according to equation (1), the maximum turbulence intensity and the integral length scale at an axial distance of $x/d \approx 7$ were used. The reasons for this choice will be given later.

EXPERIMENTAL SET-UP

The experiments were performed in a high-pressure combustion test rig that is designed for a maximum operating pressure of 30 bars and a maximum airflow rate of 750 m_N^3/h. The combustion air can be preheated up to 823 K with an electrical heater. The fuel (methane) and the combustion air are mixed homogeneously and fed into a generic combustor, which is shown in Figure 2.

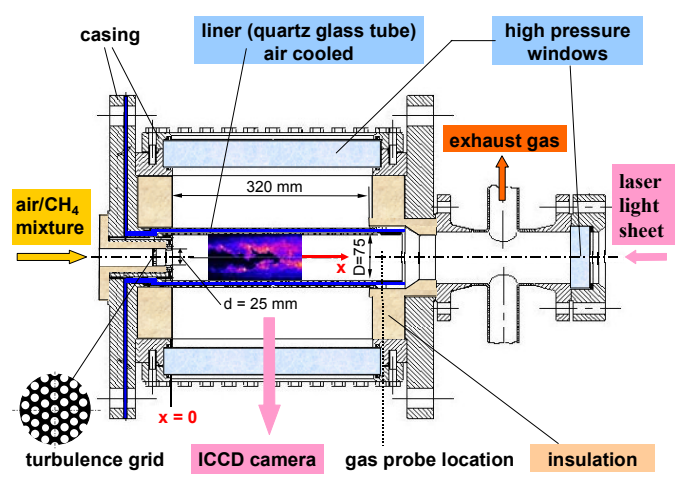

Figure 2: High-pressure combustor

A hydrogen torch igniter is used to ignite the fuel/air mixture. The flame is stabilized aerodynamically via the recirculation of hot flue gases, induced by the combustor geometry with sudden expansion. The liner consists of two coaxial quartz glass tubes, being convectively cooled with cooling air. The diameter of the inner quartz glass cylinder D is 75 mm. The diameter of the combustor inlet d is 25 mm. The combustor is especially

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designed to achieve an excellent optical access for the use of non-intrusive laser diagnostics. Therefore three large high-pressure windows, two in the sidewalls and one at the exit, were implemented in the water-cooled combustor casing. With the aid of a water-cooled gas probe located at the exit of the combustor and conventional exhaust gas analysis, the concentration of the main species (CO, CO_2, O_2, NO_x, UHC) can be measured. The hot exhaust gases exit the combustor via a water-cooled pressure throttle and are then transported into the chimney. The combustor with the maximum thermal power of 400 kW can be operated up to exhaust gas temperatures of 1850 K.

The turbulence parameters u' and L_T are varied by using different turbulence grids (hole diameter d_H, blockage) or by mounting these grids at different axial positions within the combustor inlet section (see Figure 2). In this study perforated plates with hexagonally located holes, two different blockage ratios and two different grid positions ($x_g = 10$ and 30) were used. The nomenclature of the grids is explained in Table 1. The variable x_g is defined as the axial distance from the grid position to the combustor inlet (x = 0 mm), normalized by the grid hole diameter d_H. Grid position x_g10 means e.g. the grid was mounted 10 hole diameters upstream of the combustor head.

Grid nomenclature	grid hole diameter d_H [mm]	blockage ratio [%]
g350	3	50
g365	3	65
g450	4	50
g465	4	65

Table 1: Grid nomenclature

The experimental conditions and the relevant parameters for the studied flames are presented in Table 2 and in Figure 1.

	Isothermal experiments	Combustion experiments
Bulk velocity [m/s]	20 - 40	40 – 80
Inlet temperature [K]	293	673
Pressure [bar]	1	1- 10
Equivalence ratio	-	0.5
Reynolds number $Re_d = (u_{bulk} \cdot d)/\nu$	$3.2 \cdot 10^4 – 6.4 \cdot 10^4$	$2.4 \cdot 10^4 – 2.5 \cdot 10^5$
Turbulence intensity u' [m/s] at x/d = 7	4 – 8	8– 16
Integral length scale L_T [mm] at x/d = 7	10	10
Laminar flame speed S_L [m/s]	-	0.15 – 0.63
Laminar Zeldovich thickness δ_L	-	46 - 112
Turbulent Reynolds number Re_T	-	1160 – 12460
Damköhler number Da	-	3 - 7
Karlovitz number Ka	-	5 - 30
x-Coordinate Borghi L_T/δ_L	-	90 - 218
y-Coordinate Borghi u'/S_L	-	13 - 58

Table 2: Experimental conditions and relevant parameters (turbulence quantities measured at x/d = 7)

The set-up used in the Planar Laser-Induced Fluorescence (OH-PLIF) measurements had been already described in [9]; therefore no description is given in the present paper.

PIV Set-up and Post Processing

The velocity field was measured with the help of 2D Particle Image Velocimetry (PIV). The laser beam generated by a double pulse Twin-YAG laser (Quantel) passed a set of lenses creating a 75 mm high light sheet, which was coupled through the axial quartz window. The glass tube was shielded with black screens to lower the amount of light reflections. All PIV measurements were performed in isothermal flow. The images were acquired with a 12 bit camera (PCO) placed on a translation stage. The flow was seeded with aluminum oxide Al_2O_3 solid particles generated in a specially designed seeding generator. The primary particle size was in the range of 1-3 µm.

The quality of the 12 bit images was enhanced in the SensiCam software and dumped into 8 bit bitmaps for post processing using the commercial software Visiflow. A preliminary study showed that ≈ 400 images are sufficient for flow field statistics. The camera image was 72 mm high and 86 mm wide. The flow field was imaged at 4 to 5 different axial positions with an overlap of 7 – 30%. During the PIV measurements bulk velocities have been varied between 20 and 40 m/s. The images were acquired with a frequency of 3 Hz in double exposure mode giving 2 images for the cross correlation algorithm of Visiflow. The time separation of the exposures was varied between 7 and 15 microseconds according to bulk velocity.

The images were analyzed with 32×32 pixel interrogation spot size and an overlap of 75 %. These settings were found to be optimal for the desired high spatial resolution (high number of vectors in the flow field). Therefore the flow field consists of 125 vectors in span wise and 157 vectors in stream wise direction. The resulting spatial resolution is 0.55 mm. The algorithm uses Gaussian fitting of the displacement peak in the cross-correlation function, allowing for detection of sub-pixel displacements. Calculated velocities were validated by means of their magnitude from 0 to twice the bulk velocity. Vectors outside of this range were rejected and replaced with vectors interpolated from the neighboring ones. As the commercial software did not provide a satisfactory algorithm for vector validation, an additional validation method was used to obtain more reliable values of the root mean square deviation (u(RMS), v(RMS)) velocities. In the following, to simplify matters, these values will be named RMS velocities.

In Figure 3, in which a representative histogram of the axial velocity is shown, spurious vectors can be seen, which are responsible for unrealistic high values of the RMS velocities. These spurious vectors were rejected using a standard deviation iterative method described in the following part. In the first step the mean velocity, and the RMS velocity were calculated. In the second step velocities outside of a velocity range ± 3 u(RMS) (99.7 % for a normal distribution) were rejected. These filtering loops were applied twice. It was found by analyzing the influence of the filtering factor on the axial RMS velocity that a filtering factor of 3×u(RMS), equal for both loops, was the best. Usually the percentage of rejected vectors due to this

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filtering was lower than 20 %. Therefore the influence on the mean velocity was very small, usually lower than 5 % (see also Figure 3).

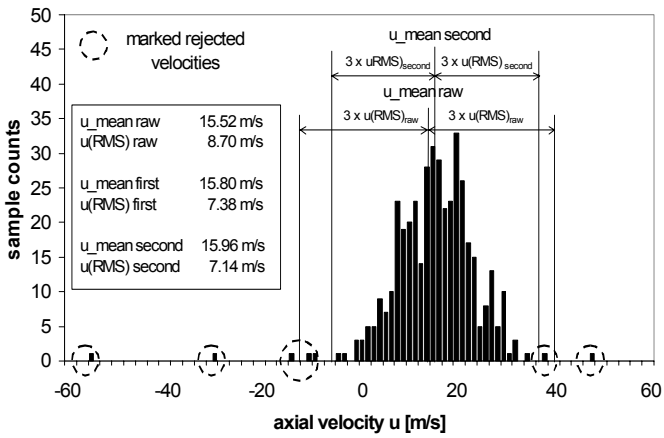

Figure 3: Histogram of the axial velocity u at the position x =164 mm, y = 5.5 mm and filtering method

The instantaneous velocity field data gives an opportunity to calculate the integral length scale L_T. These calculations were done numerically according to equations (3) and (4), and only values for the integral length scale in the axial direction will be presented in the following. A maximal correlation length r_{max} of 30 vectors (16.5 mm) was used in the calculation. The correlation coefficient R_L was plotted versus the x-coordinate and the raw data were fitted with a 6th order polynomial. According to equation (4), L_T is theoretically the integral of the correlation coefficient from 0 to infinity, but in the calculations the point at which the correlation curve crosses 0 was used to determine L_T.

RESULTS

Flow Field

The mean flow patterns are explained with the help of Figure 4, which shows the axial velocity (mean) together with streamlines and the turbulence intensity. These results of the g350 grid at the grid position x_g10, for an inlet temperature of 293 K, atmospheric pressure, and a bulk velocity of 40 m/s are representative for other grids and other operating conditions. The maximum values correspond always to the red color. In the most upstream position of the camera, close to the entrance of the combustor the laser light sheet had to be restricted in height to get rid of the reflections from the combustor head.

The *core flow region* (1) with high axial velocities and relatively low turbulence intensities can be seen clearly in Figure 4. This region, with a shape like a frustum, is located close to the axis of symmetry with an extension up to an axial distance of x/d ≈ 3. Due to the geometry with sudden expansion, a *recirculation zone* (2) is created. At an axial distance of x/d ≈ 8 the flow reattaches. Between the core flow and the recirculation zone turbulence is generated due to high velocity gradients. Close to the combustor entrance this *shear layer* zone (3) with high turbulence intensities has an annular shape and its radial extension is small. The zone spreads in radial direction with increasing axial distance x/d and at x/d ≈ 5, the high turbulent zone extends almost over the whole combustor diameter. This zone with high turbulence intensities almost over the whole diameter is named *high turbulent zone* (4) and it is about 3 diameters d long. For higher axial distances (x/d ≳ 8), towards the combustor exit, the turbulence intensity decreases due to dissipation. This region is named *dissipation zone* (5).

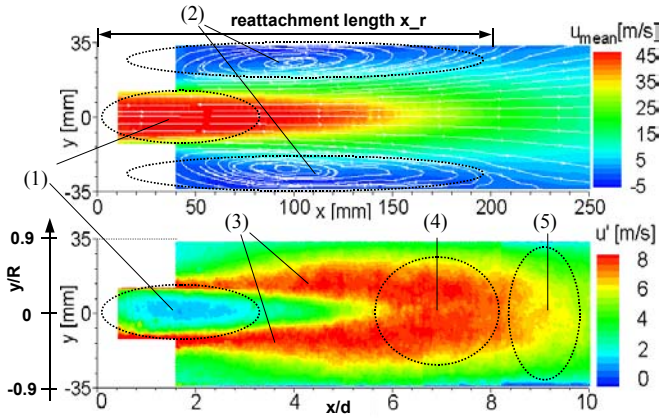

Figure 4: Axial velocity u_{mean} and turbulence intensity u' for the g350 grid (x_g10, T = 293 K, 1 bar, u_{bulk}= 40 m/s)

The Figures 5 and 6 show a comparison of the axial and radial RMS velocities. In Figure 5 axial profiles along the axis of symmetry (y/R = 0) and in Figure 6 radial profiles at different axial positions are plotted. All results are again for the same grid, grid position and operating conditions as in Figure 4.

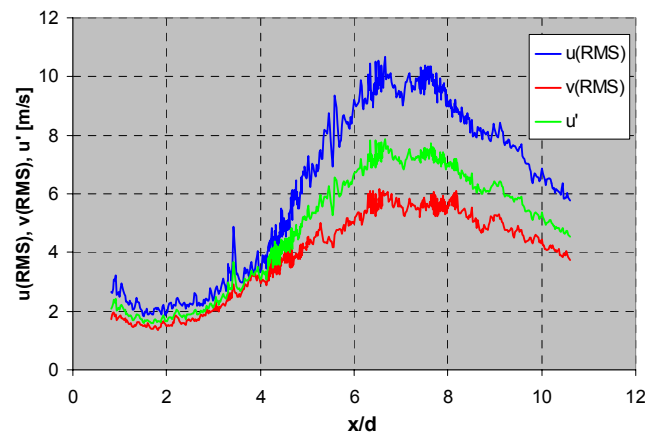

Figure 5: Axial profiles along the axis of symmetry of u(RMS), v(RMS), and u' for g350, x_g10, T = 293 K, 1 bar, u_{bulk}= 40 m/s

In the region close to the combustor head (x/d ≲ 3) the values of u(RMS), v(RMS), and u' along the axis of symmetry slightly decrease with increasing axial distance because of the dissipation of the grid-generated turbulence. Between x/d = 3 and x/d = 7 the values increase with increasing axial distance, reach their maximum at x/d ≈ 7, and decrease further downstream. This typical shape which is the same for all grids, grid positions and operation conditions is caused by the following superposed mechanisms: The generation of

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turbulence in the shear layer, the distribution of shear layer generated turbulence due to turbulent transport, and the dissipation of both, the grid and shear layer generated turbulence.

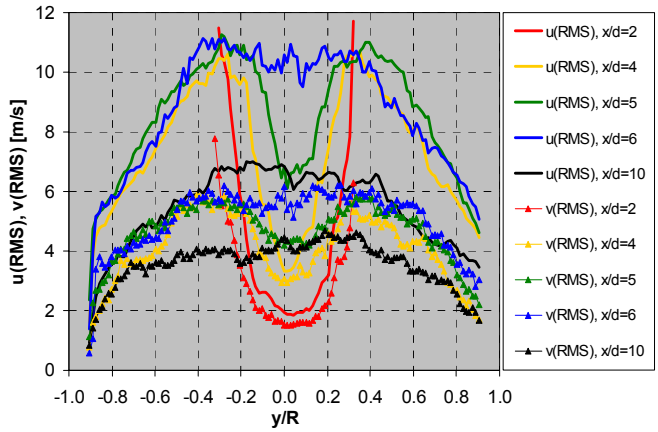

Figure 6: Radial profiles of u(RMS) and v(RMS) at different axial positions x/d for g350, x_g10, T=293 K, 1 bar, u_{bulk}= 40 m/s

The first part ($x/d \lesssim 3$) represents a region of grid-generated turbulence, in which no influence of the turbulence generated in the shear layer is present. Because the grid is mounted far upstream of this region ($x > 15\ d_H$), the dissipation rate of this grid generated turbulence is already quite small. In the second part ($3 < x/d < 7$) the values increase with increasing axial distance. Due to turbulent transport the turbulence generated in the shear layer is increasingly influencing the turbulence intensity in the region close to the axis of symmetry. This broadening of the shear layer in radial direction can be seen in the radial profiles of u(RMS) for x/d up to 5 in Figure 6. At the axial position of $x/d \approx 7$ the values of u(RMS), v(RMS), and u' reach their maximum, indicating the balance of shear layer generated turbulence, turbulent transport, and dissipation of grid and shear layer turbulence. Farther downstream ($x/d > 8$), the dissipation dominates and therefore the values of u(RMS), v(RMS), and u' decrease.

In the core flow region ($x/d \lesssim 3$, $-0.15 \lesssim y/R \lesssim 0.15$) the axial RMS velocity is only approximately 20-30 % higher than the radial RMS velocity. Therefore the turbulence in the core region is almost isotropic. This is not true for the shear layer, in which the axial RMS velocities are approximately 40 - 50 % higher than the radial RMS velocities v(RMS).

At the beginning of the PIV campaign, measurements at a higher inlet temperature of 673 K and at higher pressures (up to 5 bars) had been performed, too. Figure 7 shows the axial RMS velocity u(RMS) at the axis of symmetry for a constant bulk velocity of 20 m/s and two different inlet temperatures (293 K and 673 K), and for a bulk velocity of 40 m/s and three different pressures (1, 2, 5 bars). No influences of inlet temperature and almost no influence of pressure on the axial RMS velocity have been found. Therefore further measurements have been performed at atmospheric conditions. The higher deviation in the pressure variation series, e.g. at 5 bar, are probably due to a lower accuracy of the PIV measurements at higher densities.

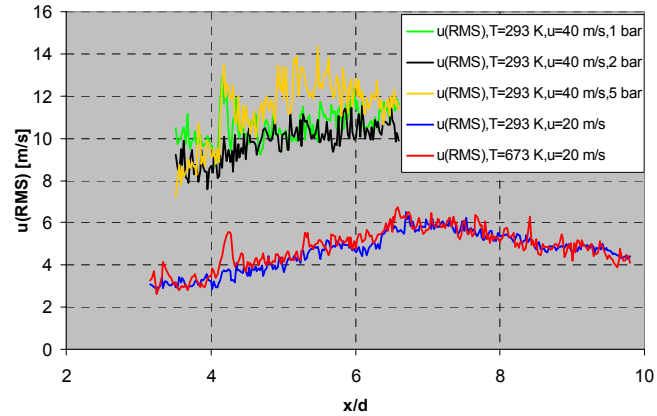

Figure 7: Axial profiles of u(RMS) along the axis of symmetry for different inlet temperatures (293 K, 673 K), pressures (1, 2, 5 bars), and two bulk velocities (20 m/s, 40 m/s) for the g350, x_g10 grid

Figure 8 shows profiles along the axis of symmetry of the mean axial velocity u_{mean} and the turbulence intensity u' of the g350 and g450 grid, for 1 bar, 293 K, and the grid position $x_g = 10$. The bulk velocity, calculated with the air flow rate and the inlet cross section, has been varied from 20 m/s to 40 m/s. For axial distances up to $x/d \approx 3$ the values of u_{mean} for both grids are constant for a constant airflow rate. The values at the axis of symmetry are approximately 30 % higher than the calculated bulk velocities. This is consistent with the theory of partly developed turbulent flows in pipes [10]. Because of the increasing influence of the expansion farther downstream, the axial velocities decrease. At higher bulk velocities due to higher flow rates, the turbulence intensity is higher. Almost no difference can be seen between the two different grids g350 and g450.

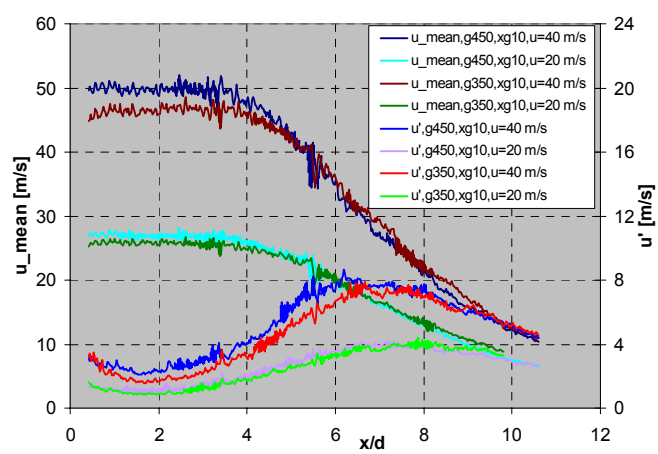

Figure 8: u_{mean} and u' along the axis of symmetry for two different grids (g350, g450) and different bulk velocities (20 m/s, 40 m/s); (T =273 K, 1 bar, x_g10)

The results of the grid position variation are shown in Figure 9, in which the turbulence intensity and the quotient u'/u_{bulk},

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named relative turbulence intensity, of the g450 grid are plotted for bulk velocities of 20 m/s and 30 m/s and two different grid positions (x_g10 and x_g30). Up to an axial distance of $x/d \approx 3$ (core flow), where the grid-generated turbulence is dominant, the values of u' for the x_g10 series are 20-30 % higher than the x_g30 ones. Because the grid in the x_g30 series is mounted more upstream than in the x_g10 series, the grid-generated turbulence has been already more dissipated. Farther downstream where the influence of the grid-generated turbulence is low in comparison with the shear layer generated turbulence, no differences between the x_g10 and x_g30 series can be observed. For different flow rates (bulk velocities), the relative turbulence intensity u'/u_{bulk} along the axis of symmetry is the same. This indicates the proportionality of the turbulence intensity to the flow rate, which can also be seen in Table 3, in which the results of the velocity variation are summarized for all grids and grid positions.

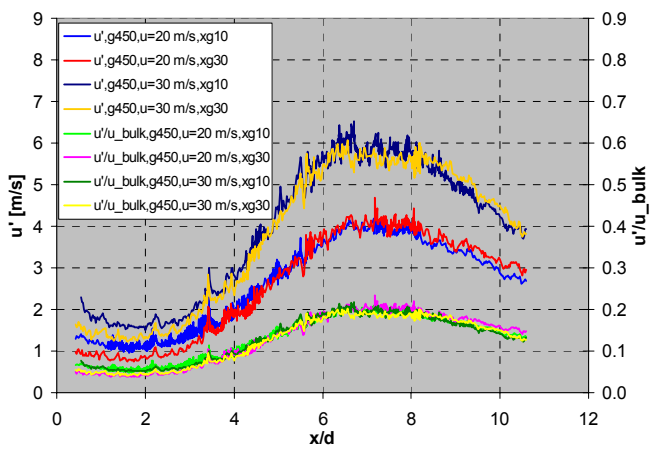

Figure 9: Turbulence intensity u' and relative turbulence inetensity u'/u_{bulk} along the axis of symmetry for the g450 grid and two different grid positions (x_g10, x_g30) and bulk velocities (20 m/s, 30 m/s); (T = 293 K, 1 bar)

In Table 3 the maximum values for the relative turbulence intensity u'/u_{bulk} at the axis of symmetry, the relative turbulence intensity u'/u_{bulk} at $x/d = 2$, and the normalized reattachment length x_r/d are summarized for all grids, grid positions and bulk velocities. For a constant grid position, the maximum values of u'/u_{bulk} and the values at $x/d = 2$ are approximately constant for each grid and for different bulk velocities, thereby indicating the proportionality of the relative turbulence intensity to the flow rate.

The maximum values of the relative turbulence intensity, reached at $x/d \approx 7$, are for all grids in the range of 20 % and do not depend on grid position or blockage ratio, because at the axial distance, where the values reach their maximum, the turbulence is dominated by the shear layer generated turbulence and not by the grid-generated turbulence.

For the core flow region, here represented by the values at $x/d = 2$, mounting the grid further upstream (x_g30 instead of x_g10) results in a decrease of the relative turbulence intensity. Because of the more upstream position of the grid the grid-generated turbulence is already dissipated to a larger extend. The value of the g465, x_g30 grid is probably too high, but no explanation can be given so far. For grids with a lower blockage ratio (g350, g450) the relative turbulence intensity in the core flow is lower than for the grids with higher blockage ratio (g365, g465). The relative turbulence intensities at $x/d = 2$ for the g350 and g450 grid are about 4 - 6 %, whereas for the g365 and g465 grid the values are in the range of 6 - 11 %. Grids with a higher blockage ratio generate more turbulence in the region close to the grid. Farther downstream, where only a part of this turbulence has been dissipated, this results in higher turbulence levels.

The normalized reattachment length x_r/d are for all other grids in the range of 8 ± 10 % and do not depend on the airflow rate (bulk velocity), which is consistent with literature data [11]. The deviation of the reattachment length of ± 10 % is probably within the accuracy limit of the presented PIV measurements.

grid	u_bulk [m/s]	grid position	u'_{max}/u_{bulk} at y/R=0 [%]	u'/u_bulk at y/R=0, x/d = 2 [%]	reattach-ment length x_r/d
g350	20	xg10	21.8	4.4	8.2
		xg30	21.9	4.0	8.2
	40	xg10	19.7	4.2	8.2
g365	20	xg10	23.4	8.2	8.4
		xg30	21.9	5.3	8.6
	40	xg10	24.6	10.5	7.8
g450	20	xg10	21.0	5.9	8.1
		xg30	23.4	4.4	8.6
	30	xg10	21.7	5.0	8.4
		xg30	20.3	4.4	8.6
	40	xg10	21.6	5.8	8.0
g465	20	xg10	19.4	6.5	7.6
		xg30	22.1	8.3	7.2
	30	xg10	19.0	6.0	7.4

Table 3: Comparison of the maximum relative turbulence intensity u'/u_{bulk} at the axis of symmetry, u'/u_{bulk} at $x/d = 2$, and the normalized reattachment length x_r/d for all grids, grid positions and different bulk velocities (T = 293 K, 1 bar).

Length Scale

The Figures 10 and 11 show results of the integral length scale along the axis of symmetry. In Figure 11 the influence of the grid hole diameter is presented for two grids (g450, g350) with a blockage ratio of 50 %, for a bulk velocity of 20 m/s, and for the grid position x_g10.

At the combustor head at $x/d = 0$, the values of the integral length scale L_T are in the range of one half of the grid hole diameter. With increasing axial distances x/d the values increase because of the preferred dissipation of small eddies,

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thereby shifting the turbulence spectrum towards bigger scales. In the core flow region (x/d ≲ 3), the integral length scales of the g450 series are about 30 % higher than for the g350 series, which is in the range of the grid hole diameter difference. This difference diminishes farther downstream, because the turbulence is increasingly influenced by the turbulence generated in the shear layer.

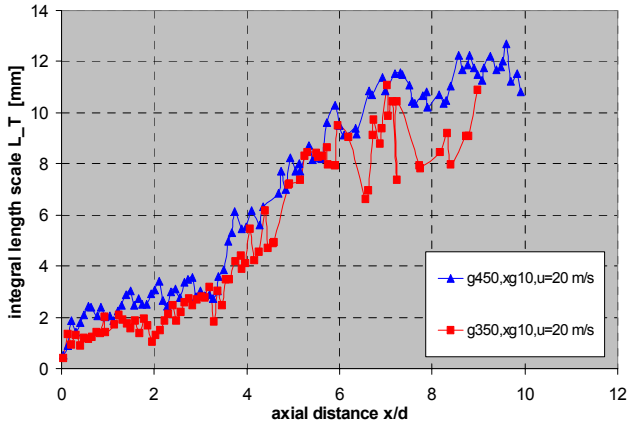

Figure 10: Influence of grid hole diameter on integral length scale L_T along the axis of symmetry for the g450 and the g350 grid (T = 293 K, 1 bar, x_g10, u_{bulk} = 20 m/s)

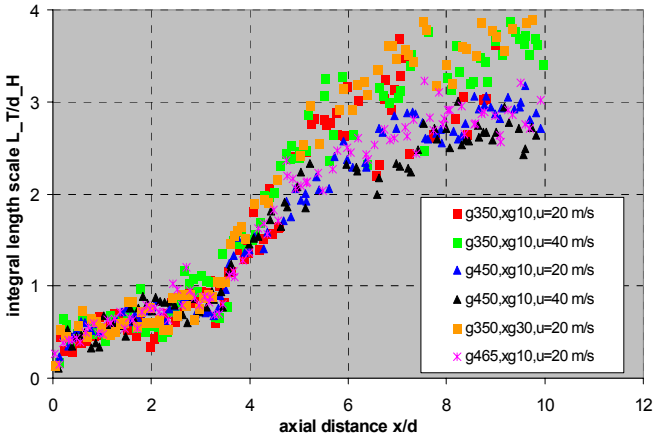

Figure 11: Normalized integral length scale L_T/d_H along the axis of symmetry for different hole diameters, blockage ratios, grid positions, and bulk velocities (T = 293 K, 1 bar)

Therefore three different regions can be distinguished: The first region of grid-generated turbulence, in which turbulence intensity and integral length scale are both being influenced by the grid geometry. This region is located close to the combustor head (0 ≲ x/d ≲ 3). In the second region farther downstream (3 ≲ x/d ≲ 7) the increase of the integral length scale with increasing axial distance is higher than in the first region. In the third zone (x/d ≳ 7) the increase of the integral length scale is very small. This typical behavior, which is even more obvious in Figure 11, is for all measurements the same. Figure 11 shows the normalized integral length scale L_T/d_H along the axis of symmetry for a variation of hole diameter, blockage ratio, bulk velocity, and grid position. All variations show these typical trends. The gradient of the integral length scale in the core flow region is lower than in the region 3 ≲ x/d ≲ 7. Due to the decrease of the dissipation rate with increasing axial distance, the increase of L_T in the core region is small, because most of the grid-generated turbulence has already been dissipated. This is different in the region (3 ≲ x/d ≲ 7), where the influence of shear layer generated turbulence due to turbulent transport is growing and the dissipation rate is high. Therefore the increase of the integral length scale is also high. Farther downstream (x/d ≳ 7), because of the long axial distance to the turbulence generation region, the dissipation rate is again low, causing the smaller increase of the integral length scale.

Flame Structure

In the following part some results of the mean flame front position and the flame front structure study for the g350, xg10 grid are presented. The following OH-PLIF images are representative for a systematic variation of the characteristic parameters Re_T, *Da, Ka* by changing the pressure and bulk velocity. More results can be found in [9]. The display window shown in the following figures corresponds in the radial direction to the full liner diameter and in the axial direction to 126 mm. In the case of some averaged images two sets of images at two axial positions has been assembled with an overlap. The averaged images are always located on the left side in the figures and the single shots on the right side. Plotted is the fluorescence intensity, which is a measure of the relative concentration of the OH radical. The yellow color corresponds always to the maximum intensity.

In Figure 12 the results of the OH-PLIF measurements for an increase of the turbulence intensity by changing the bulk velocity of the air/fuel mixture for an inlet temperature of 673 K, a pressure of 1 bar and an equivalence ratio of 0.5 are shown. In the averaged images on the left side, the same fluorescence intensity scale is used, whereas each single shot is scaled individually. Increasing the bulk velocity and thereby the turbulence intensity increasingly corrugates the flame front but does not affect the mean flame front position. The region with the black color, located upstream close to the axis of symmetry is a region of fresh gas, where no OH radicals can be found. The beginning of the mean flame front at the axis of symmetry is marked in the images with a white vertical line. This position, named *x_flame*, has been determined as follows: The intersection between the axis of symmetry and the tangent at the inflection point of the axial intensity profile at y/R = 0 is used to determine the beginning of the mean flame front. A more detailed description is given in [9]. For the calculation of the Re_T, *Da, Ka* a relative turbulence intensity u' of 20 % and an integral length scale L_T of 10 mm have been used. These are the values on the axis of symmetry at x/d = 7 adjacent to the flame front (see Figure 14).

The results of the pressure variation series at a constant inlet temperature of 673 K, a bulk velocity of 45 m/s and an equivalence ratio of 0.5 are presented in Figure 13.

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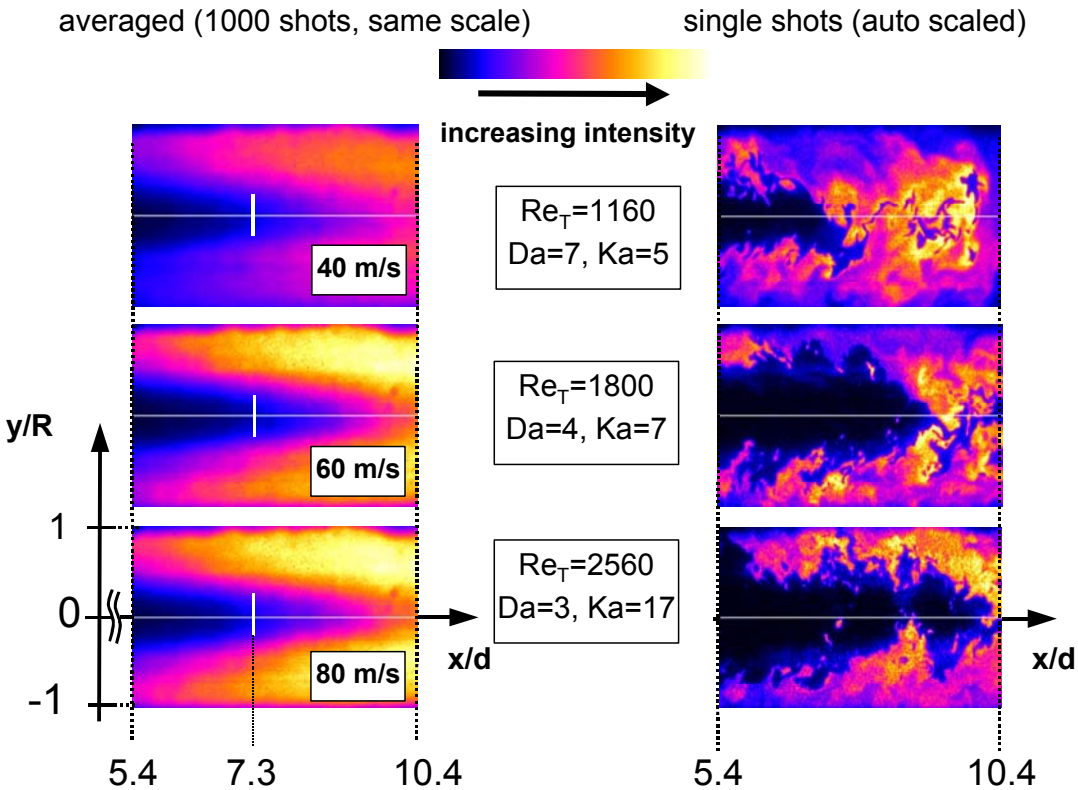

Figure 12: Fluorescence intensity of the OH radical for the g350, xg10 grid and a bulk velocity variation from 40 m/s to 80 m/s (T=673 K, 1 bar, Φ=0.5)

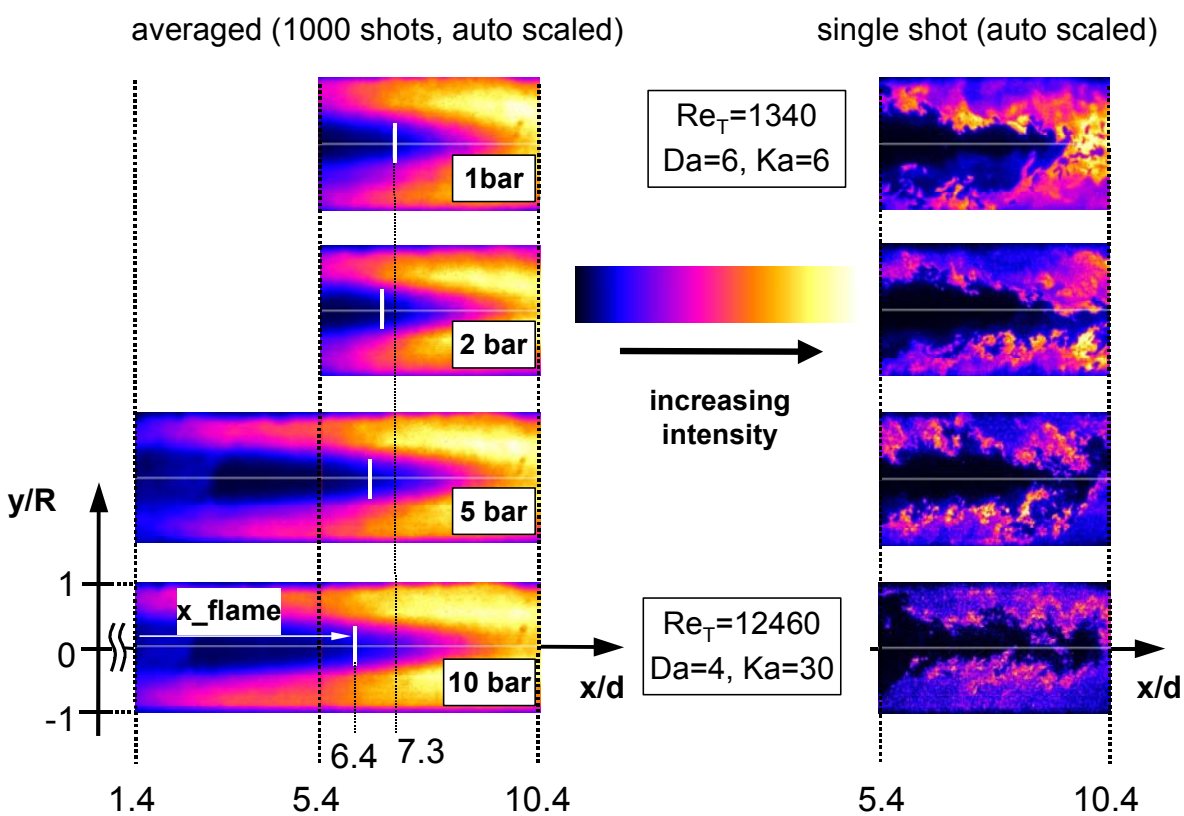

Figure 13: Fluorescence intensity of the OH radical for the g350, xg10 grid and a pressure variation from 1 to 10 bars (T=673 K, u_{bulk} = 45 m/s, Φ=0.5)

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Increasing the pressure from 1 to 10 bars and thereby increasing the turbulent Reynolds number by about one order of magnitude up to 12460 and increasing the Karlovitz number by a factor of about 5 from 6 up to 30, results in a 60 % decrease of Damköhler number down to 4 (see also Figure 1). Increasing the pressure results in a slight shift of the mean flame front position upstream towards the combustor inlet, which could be interpreted as a small increase in turbulent flame speed. The flame front is thereby increasingly corrugated, which can be seen in the single shot images on the right side of Figure 13. The value of mean flame front position at the axis of symmetry at 10 bars is about 12 % smaller than the value at 1 bar. The reason is probably slightly higher flame temperatures at higher pressures (higher thermal loading) due to lower relative heat losses. Therefore, especially in consideration of the small change in flame temperature, no significant influence of pressure on mean flame front position has been observed. This indicates that the turbulent flame speed is independent of pressure, which is contrary to laminar flames, which show a strongly decreasing laminar flame speed with increasing pressure.

Finally the reason for the use of the specific values of the relative turbulence intensity $u'/u_{bulk} = 20$ % and the integral length scale $L_T = 10$ mm has to be given. This is elucidated with the help of Figure 14, which shows the turbulence intensity together with the contour of the mean flame front for the g350, x_g10 grid, for T = 673 K, 1 bar, and a bulk velocity of 40 m/s.

The flame front contour has been derived by the determination of the flame front position at the axis of symmetry according to the previous description. The same OH-LIF signal intensity is used to obtain the whole flame front contour in the averaged OH-PLIF image. Figure 14 shows, that the flame front is located where the turbulence intensity reaches its maximum. At the axis of symmetry this is at the axial distance of x/d ≈ 7. Therefore the values of u' and L_T at this axial position had been used for the calculation of Re_T, Da, Ka, which is at least for classification purposes in the Borghi diagram a valid approach.

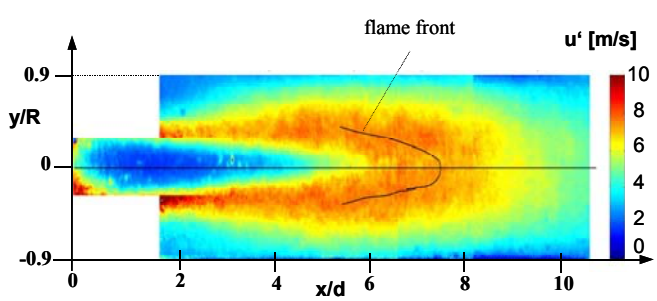

Figure 14: Turbulence intensity u' and superposed mean flame front for the g350, xg10 grid at a bulk velocity of 40 m/s (T=673 K, 1 bar, Φ=0.5)

CONCLUSIONS

In the present experimental work turbulent premixed methane/air flames have been studied at gas turbine relevant conditions. The influence of turbulence grid parameters (geometry, grid position) as well as of the operating conditions (bulk velocity, temperature, and pressure) on turbulence intensity and integral length scale have been investigated at isothermal conditions with the help of Particle Image Velocimetry (PIV). Combustion experiments with a systematic variation of the characteristic parameters, turbulent Reynolds number, Damköhler number, and Karlovitz number, by changing the pressure and bulk velocity have been performed to study the mean flame front position and the flame front structure with the help of planar Laser-Induced Fluorescence of the OH radical (OH-PLIF).

The presented PIV results show that the turbulence intensity is proportional to the bulk velocity, with a maximum value u'/u_{bulk} of 20 % reached at an axial distance of x/d ≈ 7. In the core flow region close to the combustor head, the turbulence is influenced by the grid geometry and position. Grids with higher blockage ratios or grids, mounted farther downstream, result in higher turbulence intensities in the core flow region. Farther downstream the turbulence generated in the shear layer is dominant and no influence of the grid geometry or the grid position have been observed. These results of the turbulence intensity measurements can also be assigned to the integral length scale results. Because of a low dissipation rate in the core flow region the integral length scale increases only moderately, whereas in the region farther downstream, where the influence of the shear layer turbulence due to turbulent transport is growing and the dissipation rate is high, the integral length scale increases more rapidly. At the combustor entrance the values of the integral length scale correspond approximately with one half of the according grid hole diameter d_H. Farther downstream, at the position on the axis of symmetry where the turbulence intensity is maximal, the integral length scale reaches a value of about 10 mm.

Results of the OH-PLIF measurements show that increasing the pressure and thereby primarily increasing the turbulent Reynolds number shifts the mean flame front position only slightly towards the combustor inlet, but increasingly corrugates the flame front. Increasing the bulk velocity does not affect the mean flame font position, but due to higher turbulence intensities increasingly corrugates the flame front. The comparison of the PIV and the OH-PLIF results show that the flame stabilizes in a region where the turbulence intensity is maximal.

Future studies will focus on the statistical flame front analysis to determine turbulent flame speeds. The influence of turbulence on the turbulent flame speed and flame front structure will be further investigated in the future work.

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